

YATHRA GO

A Smart Metro Bus Transportation Management System

Higher National Diploma in Information Systems Agile and DevOps 2025.2F

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1. Introduction

Public transportation is essential need of society where it provides affordable and accessible mobility for many people every day. As urban populations is continuously growing, therefore the demand for efficient, reliable and safe public transport systems has significantly increased. Many existing public transportation systems continue to face operational and service- related challenges that negatively affect passenger experience.

Although digital metro bus applications have been introduced to assist commuters with route and timetable information, most of these applications provide only basic functionalities. But they often lack real-time passenger information, intelligent route recommendations, comprehensive safety mechanisms, and advanced operational support for transport authorities. As a unexpected delays, safety concerns, limited accessibility and difficulties in making informed travel decisions.

To overcome these challenges, this project introduces YathraGo, where it is an enhanced and efficient smart metro bus application that extends the capabilities of the existing metro bus system through innovative digital technologies.

The proposed solution integrates real-time GPS-tracking, live bus occupancy monitoring, weather-aware route recommendations, smart safety monitoring, digital payment facilities, accessibility support and intelligent passenger service into a single platform.

Project's primary objective of YathraGo is to improve and provide passenger convenience, safety and satisfaction while enhancing the operational efficiency of public transportation providers. This project identified the gaps in existing services and provided valuable suggestions. The project includes the Scrum Agile methodology which is more suitable, enabling iterative development, continuous stakeholder feedback and incremental delivery of valuable system features throughout the software development lifecycle.

2. Business problem

Public transportation has advanced technological advancement which is appreciable. Although existing metro buses provide useful features for passengers, there are numerous challenges faced by passengers and transport authorities which reduce the service quality and satisfaction of the passengers.

Problems faced by passengers:

- Overcrowding is a major issue. The main reason is that passengers are unable to view the bus occupancy level before boarding and everyone takes the same crowded bus to travel, whereas some passengers wait for so long at the bus stop to get a less crowded bus. This causes inconvenience for passengers.
- Inaccurate predictions caused by traffic and other problems may reduce the satisfaction of the passengers.
- One critical concern is passenger safety, where many accidents take place. Existing system lacks support for emergency situations like harassment, accidents, or medical emergencies. Absence of quick emergency response features.
- There is only limited support for people with disabilities where there is no voice assistance and accessibility features that support them and give a comfortable journey.
- Weather related interruption is also a common problem that is faced by many passengers. Heavy rain, flooding, landslides and other disasters cause delays in service and passengers are not aware of it, they receive only little information about delays and alternative routes.
- Passengers face poor user experience. Existing applications provide only the maps with limited intelligent features, which are not enough for the passengers and there is no integrated platform providing real time information, digital services and safety.

Problems faced by transport Authority:

- Operational Monitoring is limited to such difficulty in monitoring vehicle performance in real time, it is unable to track driver behavior, visibility of service disruption is limited and there is no way to monitor passenger demand.
- Unable to monitor safety where there is no system to detect accidents automatically. Coordinating emergency response is limited.
- Lack of analytical tools, inefficient allocation of resources reduces operational efficiency. Mainly, the existing system doesn't include Data drive decision making.
- Service management problems are faced by transport authorities more often. Difficulty in managing overcrowding because there is no way to monitor real time occupancy, lack in providing an accurate arrival time and difficulty in communicating with passengers about the weather disruption and alternative routes.

3. Proposed solution

To overcome the identified challenges this project proposes YathraGo, an intelligent metro bus system through advanced smart transportation technologies. Rather than replacing the current metro bus application, YathraGo extends its functionality by introducing innovative features that improve both passenger experience and transport management.

The proposed solution includes the following key features:

Real-Time Bus Information

- Live bus occupancy monitoring.
- Crowd level indicators (Low, Medium, High).
- GPS-based bus tracking.
- Accurate Estimated Time of Arrival (ETA).
- Personalized travel recommendations.

Weather-Based Smart Routing

- Real-time weather monitoring.
- Weather alerts for rain and flooding.
- Safer alternative route suggestions during adverse weather conditions.

Passenger Safety Features

- SOS emergency assistance.
- Harassment reporting.
- Driver speed monitoring.
- Dangerous driving detection.
- Automatic accident detection.
- Emergency response activation

Accessibility Support

- Accessibility information for passengers with disabilities.
- Voice-assisted navigation.
- Assistance request facilities.

Cashless Fare Payment

- Digital wallet integration.
- QR-code ticketing.

- Secure online payment processing.

Transport Management

- Fleet operation monitoring.
- Driver performance monitoring.
- Safety incident management.
- Passenger demand analysis.
- Weather disruption monitoring.
- Service quality monitoring.
- Centralized dashboards and analytical reports.
- Intelligent recommendations for resource allocation and operational decision-making.

Overall, the proposed YathraGo system aims to modernize bus transportation by improving operational efficiency, enhancing passenger satisfaction, and supporting transport authorities with real-time information and intelligent decision-support capabilities and provide quality products where it is the solution for current transportation problem.

4. Scrum team roles

The YathraGo project follows the Scrum Agile framework, which promotes iterative software development. Scrum supports the project team to deliver functional software in incremental releases while maintaining high product quality and ensuring that business value is continuously delivered and increasing customer satisfaction. The framework encourages continuous stakeholder collaboration and rapid adaptation to changing requirements. The team improves product quality and ensures that the developed solution effectively addresses real-world transportation challenges.

Product Owner

The Product Owner represented the stakeholders and was responsible for ensuring that YathraGo addressed the needs of both passengers and transport authorities. Continuous collaboration with stakeholders and the development team ensures that the final system aligns with user needs, achieves the project's objectives, and delivers maximum business value.

Responsibilities:

- Defining the vision and objectives of the YathraGo project.
- Gathering and analyzing business and user requirements.
- Creating and prioritizing the Product Backlog.
- Prioritizing key features based on business value, including:
- Real-Time Bus Capacity Monitoring, GPS based bus tracking, ETA, passenger safety features, smart speed monitoring and accident detection, weather based smart routing, digital fare payments and accessibility Support
- Reviewing completed sprint deliverables and providing feedback for future improvements.
- Ensuring that each sprint delivered maximum value to stakeholders.

Scrum Master

The Scrum Master ensured that the Scrum framework was effectively implemented throughout the YathraGo project and this role facilitates Agile practices, supported collaboration and communication among team members, sprint progress monitoring and ensures that the project was completed efficiently within the planned timelines while maintaining software quality.

Responsibilities:

- Organizing Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective meetings.
- Conducting communication and collaboration among team members.
- Monitoring sprint progress and ensuring timely completion of planned user stories.

- Identify and remove obstacles that affect project progress.
- Ensuring that Scrum principles and Agile best practices were followed throughout the project.
- Supporting continuous improvement to enhance team productivity and software quality.

Development Team

The Development Team was responsible for analysing, designing, developing, testing, and delivering the YathraGo application through iterative sprint cycles and the team collaborated throughout the software development lifecycle to transform stakeholder requirements into a functional, reliable, and high-quality product.

Responsibilities:

- Analyzed functional and non-functional requirements.
- Designing the system architecture, database, and user interfaces.
- Developing core system features of the system.
- facilitate unit testing, integration testing, and system testing.
- Identifying defects and fixing them to provide quality products.
- Implementing stakeholder feedback received during Sprint Reviews by refining existing features and incorporating approved changes into following sprints.
- Delivering a working software increment at the end of each sprint.

5. Epics, MRFs and User Stories

EPIC 1 – Real-Time Bus Capacity & Crowd Management

- **MRF 1.1 – Live Occupancy Tracking**
 - **US-001:** As a passenger, I want to view the occupancy percentage of a bus so that I can determine how crowded it is.
- **MRF 1.2 – Crowd Level Indicators**
 - **US-002:** As a passenger, I want crowd levels displayed using color indicators so that I can quickly assess crowding.
- **MRF 1.3 – Passenger Alerts**
 - **US-003:** As a passenger, I want crowd notifications so that I can choose less crowded buses.

EPIC 2 – Bus Timing & Arrival Tracking

- **MRF 2.1 – GPS-Based Bus Tracking**
 - **US-004:** As a passenger, I want to track a bus in real time so that I know its current location.
- **MRF 2.2 – ETA Prediction**
 - **US-005:** As a passenger, I want to view estimated arrival times so that I can plan my journey.
- **MRF 2.3 – Delay Notifications**
 - **US-006:** As a passenger, I want delay notifications so that I can make alternative arrangements.

EPIC 3 – Passenger Safety & Harassment Prevention

- **MRF 3.1 – SOS Emergency Button**
 - **US-007:** As a passenger, I want an SOS button so that I can request immediate assistance.
- **MRF 3.2 – Incident Reporting**
 - **US-008:** As a passenger, I want to report incidents so that corrective action can be taken.
- **MRF 3.3 – Safety Monitoring**
 - **US-009:** As an operator, I want to monitor safety incidents so that I can respond effectively.

EPIC 4 – Smart Speed Monitoring & Accident Prevention

- **MRF 4.1 – Speed Monitoring**
 - **US-010:** As an operator, I want to monitor bus speeds so that unsafe driving can be identified.
- **MRF 4.2 – Driver Alerts**
 - **US-011:** As a driver, I want speed warnings so that I can avoid violating limits.
- **MRF 4.3 – Accident Detection**
 - **US-012:** As an operator, I want accidents detected automatically so that emergency support can be activated.

EPIC 5 – Accessibility & Disability Support

- **MRF 5.1 – Accessibility Information**
 - **US-013:** As a disabled passenger, I want accessibility information so that I can choose suitable transport.
- **MRF 5.2 – Voice Assistance**
 - **US-014:** As a visually impaired passenger, I want voice assistance so that I can navigate the app independently.
- **MRF 5.3 – Accessibility Booking Support**
 - **US-015:** As a disabled passenger, I want assistance requests so that support can be arranged before travel.

EPIC 6 – Student & University Discount System

- **MRF 6.1 – Student Verification**
 - **US-016:** As a student, I want my identity verified so that I can receive discounts.
- **MRF 6.2 – Discount Eligibility**
 - **US-017:** As a student, I want my eligibility assessed so that discounts are applied correctly.
- **MRF 6.3 – Discounted Ticketing**
 - **US-018:** As a student, I want discounted fares so that I can reduce transportation expenses.

EPIC 7 – Weather-Based Smart Routing & Alerts

- **MRF 7.1 – Weather Monitoring**
 - **US-019:** As a passenger, I want weather information so that I can plan my travel safely.

- **MRF 7.2 – Weather Alerts**

- **US-020:** As a passenger, I want weather alerts so that I can avoid travel disruptions.

- **MRF 7.3 – Alternative Route Suggestions**

- **US-021:** As a passenger, I want alternative routes suggested so that I can reach my destination despite disruptions.

EPIC 8 – Smart Queue Management System

- **MRF 8.1 – Virtual Queue**

- **US-022:** As a passenger, I want to join a virtual queue so that I can avoid physically waiting at bus stops.

- **MRF 8.2 – Boarding Predictions**

- **US-023:** As a passenger, I want boarding predictions so that I know when I can board the bus.

- **MRF 8.3 – Queue Analytics**

- **US-024:** As an operator, I want queue analytics so that I can identify congestion patterns.

EPIC 9 – Lost & Found Tracking System

- **MRF 9.1 – Lost Item Reporting**

- **US-025:** As a passenger, I want to report lost items so that I can recover my belongings.

- **MRF 9.2 – Found Item Registration**

- **US-026:** As a staff member, I want to register found items so that owners can locate them.

- **MRF 9.3 – Claim Verification**

- **US-027:** As a staff member, I want to verify ownership claims so that items are returned securely.

EPIC 10 – Feedback & Rating System

- **MRF 10.1 – Journey Rating**

- **US-028:** As a passenger, I want to rate my journey so that service quality can improve.

- **MRF 10.2 – Feedback Submission**

- **US-029:** As a passenger, I want to submit feedback so that concerns and suggestions are considered.

- **MRF 10.3 – Analytics Dashboard**

- **US-030:** As an operator, I want feedback analytics so that I can identify service improvements.

EPIC 11 – Personalized Travel Recommendation System

- **MRF 11.1 – Travel Pattern Analysis**

- **US-031:** As a passenger, I want the system to analyze my travel patterns so that recommendations can be personalized.

- **MRF 11.2 – Personalized Suggestions**

- **US-032:** As a passenger, I want personalized route suggestions so that I can optimize travel time.

- **MRF 11.3 – Smart Notifications**

- **US-033:** As a passenger, I want personalized notifications so that I receive relevant travel updates.

EPIC 12 – Demand-Based Route Optimization & Dynamic Scheduling

- **MRF 12.1 – Demand Analysis**

- **US-034:** As an operator, I want demand forecasting so that buses can be allocated efficiently.

- **MRF 12.2 – Dynamic Scheduling**

- **US-035:** As an operator, I want schedule adjustments so that services match passenger demand.

- **MRF 12.3 – Route Optimization**

- **US-036:** As an operator, I want route optimization recommendations so that operational efficiency improves.

EPIC 13 – Emergency Medical Support

- **MRF 13.1 – Emergency Medical Request**

- **US-037:** As a passenger, I want to request medical assistance so that I can receive emergency support.

- **MRF 13.2 – Emergency Contact Integration**

- **US-038:** As a passenger, I want emergency contacts notified so that family members are informed.

- **MRF 13.3 – First Aid Support**

- **US-039:** As a passenger, I want first aid guidance so that I can receive immediate assistance.

EPIC 14 – Driver Role-Based Smart Dashboard

- **MRF 14.1 – Driver Dashboard**

- **US-040:** As a driver, I want to view my assigned routes so that I can manage my trips efficiently.

- **MRF 14.2 – Performance Monitoring**

- **US-041:** As a driver, I want to monitor my performance metrics so that I can improve service quality.

- **MRF 14.3 – Vehicle Status Monitoring**

- **US-042:** As a driver, I want to monitor vehicle health so that maintenance issues can be identified early.

EPIC 15 – Driver Communication & Control Center Integration

- **MRF 15.1 – Two-Way Messaging**

- **US-043:** As a driver, I want to communicate with the control center so that I can receive operational support.

- **MRF 15.2 – Operational Alerts**

- **US-044:** As a driver, I want operational alerts so that I can respond to service changes.

- **MRF 15.3 – Escalation Management**

- **US-045:** As a control center operator, I want escalation management so that incidents can be handled efficiently.

EPIC 16 – Driver Real-Time Traffic & ETA Prediction

- **MRF 16.1 – Traffic Monitoring**

- **US-046:** As a driver, I want live traffic information so that I can avoid congestion.

- **MRF 16.2 – ETA Calculations**

- **US-047:** As a driver, I want ETA predictions so that passengers receive accurate arrival times.

- **MRF 16.3 – Route Suggestions**

- **US-048:** As a driver, I want alternative route suggestions so that I can reduce travel delays.

EPIC 17 – Driver Incident Reporting & Emergency Handling

- **MRF 17.1 – Incident Reporting**

- **US-049:** As a driver, I want to report incidents so that operational teams can respond.

- **MRF 17.2 – Emergency Escalation**

- **US-050:** As a driver, I want emergency support requests so that assistance can be dispatched quickly.

- **MRF 17.3 – Incident Tracking**

- **US-051:** As an operator, I want incident tracking so that cases can be monitored until resolution.

EPIC 18 – Driver Weather Alert & Route Adjustment

- **MRF 18.1 – Weather Monitoring**

- **US-052:** As a driver, I want route-specific weather information so that I can travel safely.

- **MRF 18.2 – Route Adjustment Support**

- **US-053:** As a driver, I want route adjustment recommendations so that I can avoid hazardous conditions.

- **MRF 18.3 – Weather Reporting**

- **US-054:** As a driver, I want to report weather conditions so that operational awareness improves.

EPIC 19 – Driver Shift Management & Scheduling

- **MRF 19.1 – Shift Assignment**

- **US-055:** As a driver, I want to view my assigned shifts so that I can plan my work schedule.

- **MRF 19.2 – Attendance Management**

- **US-056:** As a driver, I want attendance records maintained so that work hours are tracked accurately.

- **MRF 19.3 – Leave Management**

- **US-057:** As a driver, I want to request leave so that scheduling conflicts can be managed properly.

EPIC 20 – Cashless & Digital Fare Payment System

- **MRF 20.1 – Digital Wallet**

- **US-058:** As a passenger, I want a digital wallet so that I can pay fares conveniently.

- **MRF 20.2 – QR Ticketing**

- **US-059:** As a passenger, I want QR tickets so that I can board buses digitally.

- **MRF 20.3 – Payment Gateway Integration**

- **US-060:** As a passenger, I want secure online payments so that transactions can be completed safely.

EPIC 21 – Payment Processing, Refund & Revenue Management

- **MRF 21.1 – Fare Processing**

- **US-061:** As an operator, I want fares processed accurately so that revenue records remain correct.

- **MRF 21.2 – Refund Management**

- **US-062:** As a passenger, I want refund requests processed so that I can recover eligible payments.

- **MRF 21.3 – Revenue Analytics**

- **US-063:** As an operator, I want revenue analytics so that financial performance can be monitored.

6. Planning poker

User Story	Member 1	Reasoning	Member 2	Reasoning	Average
US-007: As a passenger, I want an SOS button so that I can request immediate assistance.	8	Requires emergency alert generation, GPS integration, and notification handling. Moderate complexity due to safety-critical functionality.	13	Includes real-time location sharing, emergency services integration, and reliability requirements.	10.5
US-008: As a passenger, I want to report incidents so that corrective action can be taken.	5	Basic reporting feature with form submission and evidence upload.	8	Requires image upload, database storage, and validation workflow.	6.5
US-009: As an operator, I want to monitor safety incidents so that I can respond effectively.	13	Requires dashboard development, incident tracking, and real-time monitoring.	13	Needs analytics, alerts, and operator-side management features.	13
US-010: As an operator, I want to monitor bus speeds so that unsafe driving can be identified.	13	Requires IoT/GPS speed data integration and monitoring dashboard.	8	Speed tracking is complex but mainly involves displaying collected data.	10.5
US-011: As a driver, I want speed warnings so that I can avoid violating limits.	5	Alert-based feature with predefined speed thresholds.	8	Requires real-time speed detection and driver notification mechanism.	6.5
US-012: As an operator, I want accidents detected automatically so that emergency	21	Requires advanced sensors/AI detection, automatic alerts, and emergency	21	High uncertainty due to accident detection accuracy and hardware dependency.	21

support can be activated.		response integration.			
US-016: As a student, I want my identity verified so that I can receive discounts.	8	Requires document upload, verification process, and approval workflow.	8	Moderate complexity involving user validation and data security.	8
US-017: As a student, I want my eligibility assessed so that discounts are applied correctly.	5	Requires checking eligibility rules and updating student status.	5	Mainly involves business rules and database validation.	5
US-018: As a student, I want discounted fares so that I can reduce transportation expenses.	8	Requires integration between verification, ticketing, and fare calculation.	13	Involves payment system integration and dynamic fare calculation.	10.5

User Story Prioritization we did in the class

3.1.1 M
As a passenger, I want to press an SOS button during an emergency so that I can quickly request help

appropriate action.

has been submitted so that I know help is being arranged

properly supported

3.1.2 M
As a passenger, I want my live location to be automatically shared when I trigger an SOS alert so that

my location is automatically shared

3.3-42 M
As a transport admin, I want to monitor reported safety incidents in real time so that I can coordinate responses effectively

3.1.1 M
As a driver, I want to receive Speed warning notifications when I exceed the Speed limit so that I can take corrective action.

As a driver, I want dangerous driving alerts to be clearly visible and audible so that I do not miss critical warnings

As a driver, I want the system to notify me when I exceed the speed limit so that I can reduce my speed immediately.

As a driver, I want the system to automatically detect a crash so that emergency actions can be initiated immediately

As a transport administrator, I want dangerous driving events to be logged in the system so that safety performance can be monitored and analyzed

3.1.1 M
As a driver, I want to receive alerts for sudden acceleration or harsh braking so that I can improve my driving behaviour.

Safety regulations

driver performance can be reviewed.

As a transport administrator, I want to receive notifications about repeated dangerous driving incidents so that appropriate action can be taken

As a transport administrator, I want to receive instant accident notifications so that I can coordinate emergency response efforts.

3.3-42 S
As a passenger, I want to receive safety alerts related to my route so that I can travel more safely

As a student, I want to upload my school/university ID so that I can apply for student travel discounts

so that only eligible students receive discounts.

As a student, I want to check my discount eligibility status so that I know whether I qualify for discounted fares

transport admin, I want to define discount eligibility criteria for different student categories so that discounts are applied correctly to discount users so that I receive the correct fare

I can use my discount on my carrying paper tickets before it expires

done 11/06/2020

7. Sprint backlog

The sprint backlog that contains the user stories and tasks selected for the specific sprint during planning with the development team and the product owner by selecting the highest priority product backlog items to achieve the sprint goal of the first release of Yathra Go.

Sprint Goal

Develop and implement the core functionalities for passenger safety, smart, speed monitoring and the student discount system to enhance the safety reliability, and affordability of public transportation.

Sprint Duration

Sprint 1: 2 weeks

Sprint backlog items (using JIRA)

User Story ID	User Story	Epic	Priority	Status
US-301	As a passenger, I want an SOS emergency button so that I can request immediate assistance during emergencies	Epic 3	HIGH	Completed
US-302	As a passenger, I want to report Harassment anonymously so that I feel safer using public transport	Epic 3	HIGH	Completed
US-303	As a passenger, I want to share my live journey with trusted contacts for added security.	Epic 3	MEDIUM	Completed
US-401	As a passenger, I want to receive speed alerts whenever I exceed the speed limit	Epic 4	HIGH	Completed
US-402	As a transport operator, I want to monitor bus speeds in real time so that unsafe driving can be prevented.	Epic 4	HIGH	In Progress

US- 403	As an administrator, I want accident reports to be recorded automatically so that incidents can be investigated efficiently.	Epic 4	MEDIUM	In Progress
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Sprint Tasks

- To design the passenger safety dashboard.
- To develop SOS emergency module
- Develop anonymous harassment reporting system.
- Implement live journey sharing/live location.
- Develop speed monitoring module.
- Configure speed limit alert system.
- Create accident reporting module
- Develop student verification portal
- Integrate student discount calculation
- Perform unit testing
- Conduct integration testing
- Fix bugs and UI improvements

Expected outcome

The team thought that finishing the backlog would give them a working product increment. This product increment would make things safer for passengers. It would add speed monitoring so drivers have to be accountable for what they do. The team also wanted to give students at the university some options that they can afford. The features that the team finished would be shown at the sprint review. At the sprint review the team would get feedback from stakeholders. They would add this feedback to the product backlog, for the future sprints. The sprint review seems like a thing to do but it seems like not everything was fully settled yet.

Sprint Deliverables

Passenger Safety

- SOS Emergency Button
- Anonymous Harassment Reporting
- Live Journey Sharing

Road Safety

- Real-Time Speed Monitoring
- Driver Speed Alerts
- Operator Speed Dashboard

Student Services

- Student Registration
- Student Verification
- Automatic Fare Discount System

8. Daily Scrum Process

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- Operator Speed Dashboard

Student Services

- Student Registration
- Student Verification
- Automatic Fare Discount System

9. Sprint Review

The sprint review was conducted at the end of sprint 1 to demonstrate the completed product increment and gather the feedback from each stakeholder. The goal is to evaluate whether the sprint goal has been achieved and identify improvements for the next sprints.

The sprint review turned out interactive with stakeholders evaluating the delivered functionality together. They discussed the product direction and suggested enhancements based on user needs and business objectives. It seems this made things more collaborative than a formal acceptance meeting would have been.

The product owner reviewed completed work against the sprint backlog. This helped determine if the sprint goal had been achieved and if features met the acceptance criteria that were set. During the session the team demonstrated functionalities from epic three passenger safety and harassment prevention, epic four smart speed monitoring and accident prevention, and epic six student and university discount system.

Stakeholders assessed usability functionality and overall quality while providing recommendations. Some of those ideas were for improvements and future enhancements.

The participants for the sprint Review will be as such,

1. Product Owner – Jessica Joseph
2. Scrum Master – Sachani Paldano
3. Development Team - Kirubashini Selvakumar

Stakeholder

Features Demonstrated at the sprint review by completing the functionality,

Epic 3:- Passenger Safety & Harassment Prevention.

- SOS emergency button
- Anonymous harassment reporting
- Live location sharing

Epic 4:- Smart speed monitoring and accident prevention

- Driver speed monitoring
- Speed limit alert notifications
- Basic speed dashboard

Epic 6:- Student and university discount system

- Student Registrations

- Student ID verification
- Automatic student fare discounts

Stakeholder Feedback

Add Emergency contact management for passengers
Include GPS location automatically when submitting SOS requests.
Display speed history using charts.
Notify transport operators immediately when speed violations occur
Allow universities to verify student through institutional email addresses
Improve the verification status interface for students

Product Backlog Updates

- Emergency contact management.
- Real- time police and emergency service notifications
- Driver speed analytics dashboard
- AI based over speed detection
- University email verification
- Digital student transport pass
- Safety incident history
- Speed violation reports

The sprint outcome was successfully achieved, the core features of passenger safety, smart speed monitoring, and student discount management were completed, while stakeholder suggestions were documented and prioritized for future sprints.

10.Retrospective

We held the retrospective to evaluate the teams performance, identifying the areas we needed improvement and strengthen collaboration for future sprints. The objective was to evaluate teams performance and identify successful practices then we discuss the challenges we

encountered during the yathra go project then we defines improvement actions for the next sprint.

What went well

- Strong collaboration thought sprint1.
- Daily scrum meetings kept everyone aligned.
- Safety related features were completed according to schedule.
- Student verification workflow functioned successfully.
- Clear and transparent communication

Challenges

- GPS integration for speed monitoring required additional testing.
- Emergency workflow required more validation scenerios then we expected.
- Student verification logic need addition business rules.
- Some user stories were underestimated during the sprint planning.

Improvements for sprint 2.

- Break complex user stories into smaller development tasks.
- Improve estimation accuracy using story points.
- Conduct earlier integration testing.
- Perform peer code reviews before making stories as complete
- Increase automated testing

The meeting highlighted that the team successfully delivered the planned safety and student focused features while identifying opportunities to improve the estimation, testing and system integration. The agreed improvements were inculcated into the sprint 2 planning, supporting scrumprincipals of continues improvement and enabaling the team deliver higher quality increments in future sprints.

11.DevOps benefits

The DevOps practices provides significant benefits to our YathraGo system improving by collaborating, automating, software quality and continuous delivery throughout the development lifecycle. By the operation activities and integrating the developments the team can ensure that the new features are perfectly developed, tested and deployed efficiently while the system is being reliably maintained.

These are the key main benefits to our YathraGo project by the DevOps:

1. Continuous Integration and Continuous Delivery (CI/CD)

DevOps allows continuous integration so that the developers can frequently connect the code updates into a shared repository and test the application automatically. This will reduce the integration issues and ensures that the new features such as SOS emergency services, GPS tracing, digital payments and smart route finding functionalities are delivered effectively.

2. Improved Software Quality

Automated testing practices helps identify the defects in the early stages of the development process. Unit testing, integration testing automation deployment pipelines ensure that whether YathraGo is maintaining the high reliability, performance and security while minimizing the possibility of errors in critical features such as emergency responses and the processing of the payments.

3. Better Collaboration Between the Teams

Improved communication between the developers, testers, system administrators and the stakeholders are being encouraged by the DevOps. This helps to improve coordination during the development of complex features such as real time bus tracking, accident detection, transport management dashboards and passenger safety systems.

4. Faster Delivery of the Features

Through the DevOps automation, the development team will be able to release the new system improvements and updates the system faster. This helps Agile principles by allowing the team to deliver incremental improvements of the product based on the stakeholder feedback which was received during the sprint reviews.

5. Enhancement of the Security Management

Continuous security testing and vulnerability monitoring is supported by the DevOps practices. This is a really important benefit to our YathraGo system because the system handles sensitive data and information about the passenger details, payment methods and transactions, emergency requests and student verification.

6. Scalability and Reliability

Using the cloud based deployment and automated infrastructure management allows YathraGo to scale according to the passenger demand. During the high demand periods the system can support by increasing the users meanwhile maintaining the reliability of the service performance.

As a whole, DevOps helps to improve the efficiency, quality and maintenance of the YathraGo system by allowing continuous improvement, faster delivery and a more reliable smart transportation solution.

12.Conclusion

YathraGo was introduced as a transportation solution to deal with the problems of current public transportation systems. The main goal of this project is to make traveling safer, easier and more accessible for passengers. This is done by using digital technologies.

The system has helpful features like real-time bus tracking, information about how full the bus is, SOS emergency help, smart speed monitoring, discounts for student route suggestions based on weather, digital payments and analytics for managing transport. These features help passengers make choices when traveling. They also help transport officials run services effectively.

The Scrum Agile approach helped the team build YathraGo using an adaptable process. Tasks like planning sprints, holding meetings reviewing progress and looking back at what went well or what could be improved allowed the team to keep improving the system. They also got feedback from people involved in the project along the way.

DevOps practices also played a role in the project. These practices made software testing release, working together and keeping the system reliable better. They make sure that YathraGo can be updated and improved to meet transportation needs.

In the end YathraGo shows how Agile, DevOps and smart technologies can come together to build a public transportation system. The idea behind YathraGo can help make travel safer more efficient and easier for everyone. It also helps create a way to move around in the future.

13. Contribution List

Index Number	Contributed topics
COHNDIS25.2F-005	Introduction Business problem Proposed solution Scrum team roles
COHNDIS25.2F-006	Sprint backlog Daily Scrum Process Sprint Review Retrospective
COHNDIS25.2F-011	Epics, MRFs and User Stories Planning poker DevOps benefits Conclusion